Excluding additional points in the four even sieve classes for $X_{\mathrm{ns}}^{+}(121)$

29 October 2026

Status and scope. The new calculation excludes additional points in the sieve classes $n_0 = -2, 0, 2, 4$. The argument below uses the previously reproduced finite sieve and its cusp-descent classes as input. The higher-prime local tables in that sieve have not all been regenerated. The new even-case Selmer calculations have been certified without GRH. The three odd classes $n_0 = -1, 1, 3$ remain unresolved; this is not a determination of all rational points on $X_{\mathrm{ns}}^{+}(121)$.

1 Statement and notation

Put

$$E : y^2 + y = x^3 - x^2 - 7x + 10, \qquad G = (4, 5), \qquad E(\mathbb{Q}) = \mathbb{Z}G,$$

and let $K = \mathbb{Q}(z)$, where $z^5 + z^4 - 4z^3 - 3z^2 + 3z + 1 = 0$. Put

$$s = (-z^4 - z^3 + 2z^2 + 3z + 3, -2z^4 - 4z^3 + 6z^2 + 11z + 1), \quad \alpha = x(2s).$$

Here s has order 11. The finite sieve leaves precisely the states $n \equiv n_0 \pmod{M}$, $-2 \le n_0 \le 4$, with $v_2(M) = 4$ and $v_{11}(M) = 2$. The exact integer M is in `outputs/sieve_result.json`. In each state, the cusp descent gives

$$x(nG - 3s) = \alpha + \delta_{n_0} w^{11}, \qquad \delta_{n_0} = x(n_0 G - 3s) - \alpha, \qquad w \in K. \tag{1}$$

This is the input from the preceding two reports.

Proposition 1. *Let $n_0 \in \{-2, 0, 2, 4\}$. If $n \equiv n_0 \pmod{M}$ and (1) holds for some $w \in K$, then $n = n_0$. Consequently, subject to the inherited sieve input, any additional rational point on $X_{\mathrm{ns}}^{+}(121)$ must lie in one of the three odd states.*

Set

$$L = K(\theta), \qquad \theta^3 - \theta^2 - 7\theta + 41/4 = 0, \qquad F = L(\zeta_{11}).$$

The degrees are 15 and 30. Write $\beta^2 = \alpha - \theta$, with $\beta \in L$, and $a = \delta_{n_0}/\beta^2$. In an even state the two-descent gives $v \in L$ with

$$v^2 = \alpha - \theta + \delta_{n_0} w^{11}.$$

After $X = aw$, $Y = a^5 v/\beta$, this becomes

$$C : \quad Y^2 = X^{11} + b^2, \qquad b = a^5. \tag{2}$$

Let $P_0 \in C(L)$ be the known point corresponding to n_0, so $X(P_0) = a$, and embed C in $J = \mathrm{Jac}(C)$ by $i(P) = [P - P_0]$. Write $\sigma(X, Y) = (\zeta_{11} X, Y)$ and $\lambda = 1 - \sigma$. The group $J[\lambda](F)$ is cyclic of order 11, generated by $T = [(0, b) - \infty]$.

2 The certified Selmer calculation

At each of the three primes $v \mid 11$ of F, the completion is $\mathbb{Q}_{11}(\zeta_{11})$, with residue field $\mathbb{F}_{11}$. Denote the Kummer map by $\mu_v : J(F_v) \to J(F_v)/\lambda J(F_v)$, realized by the descent functions $Y - b$ and $Y + b$. Let $\mathcal{S}$ be the image of $\mathrm{Sel}^{\lambda}(J/F)$ under the joint localization at these three places. Below we define a formal subgroup $H_{1,v}$; put

$$U = \prod_{v|11} \mu_v(H_{1,v}), \qquad T_{\mathrm{loc}} = \prod_{v|11} \langle \mu_v(T) \rangle, \qquad T_{\mathrm{glob}} = \langle (\mu_v(T))_v \rangle.$$

The computed dimensions are

n_0	-2	0	2	4
$\dim \mathrm{Sel}^{\lambda}(J/F)$	5	5	4	4
$\dim \ker(\mathrm{loc}_{11})$	0	0	0	0
$\dim(\mathcal{S} \cap U)$	0	0	0	0
$\dim(\mathcal{S} \cap T_{\mathrm{loc}})$	1	1	1	1
$\dim(\mathcal{S} \cap (U + T_{\mathrm{loc}}))$	1	1	1	1

More precisely, exact linear algebra verifies

$$U \cap T_{\mathrm{loc}} = 0, \qquad \mathcal{S} \cap (U + T_{\mathrm{loc}}) = T_{\mathrm{glob}}. \tag{3}$$

Every local $\mu_v(T)$ is nonzero, and $\dim U = \dim T_{\mathrm{loc}} = 3$. The matrices are recorded in `formal_*_data.m`; `audit_matrices.m` verifies the identities independently of the Selmer driver.

Here is how the GRH qualification from the previous report was removed. The class numbers of L and $H = \mathbb{Q}(\zeta_{11})$ are proved to be one. The extension F/K is Galois with group S_3, and L, H are fixed fields of subgroups of orders 2 and 3. Restriction and norm, with degrees prime to 11, identify the corresponding invariant 11-primary class groups with those of these subfields. Every irreducible $\mathbb{F}_{11}[S_3]$-module has invariants under at least one of these two subgroups. Hence $\mathrm{Cl}(F)[11] = 0$.

There are four primes in S_F, namely the primes above 2 and 11. Therefore $\dim F(S_F, 11) = 14+1+4 = 19$. The split descent algebra is $F \times F$, so its global Kummer space has dimension 38. Although GRH was used to discover candidate generators, the code checks that each candidate has valuations divisible by 11 outside S_F, and that the 38 candidates have independent joint local images. This certifies the full global space without GRH. The local images at 11 are generated by actual local points until their known full dimension is reached; the routine also checks precision bounds for the Kummer coordinates. The other local calculations use Magma's descent routines. Thus the displayed even-case Selmer data are unconditional.

In particular, the previous rank upper bounds are now unconditional:

$$\mathrm{rank}\, J_{-2}(L), \mathrm{rank}\, J_0(L) \le 20, \qquad \mathrm{rank}\, J_2(L), \mathrm{rank}\, J_4(L) \le 15.$$

The exclusion argument does not require these bounds to be sharp or a finite-index Mordell–Weil subgroup.

3 The local formal calculation

Fix $v \mid 11$, write $k = F_v$, $\pi = 1 - \zeta_{11}$, and normalize $v_k(\pi) = 1$. Then $v_k(11) = 10$ and $11/\pi^{10} \equiv -1 \pmod{\pi}$. In all four cases,

$$\bar{a} = 8, \qquad \bar{b}^2 = 1, \qquad \overline{Y(P_0)} \neq 0.$$

Use the proper hyperelliptic model of (2) over $\mathcal{O}_k$. Its special fibre is geometrically integral, with one cusp, and its relative identity Picard group is smooth of dimension five. Let H_m be

its formal subgroup of points congruent to zero modulo π^m. Its cotangent module has integral basis

$$\omega_j = \frac{X^{j-1}dX}{2Y}, \qquad 1 \le j \le 5.$$

The automorphism σ extends to the model and reduces to the identity. Consequently the power series for λ in formal coordinates have all coefficients divisible by π. Their linear terms have eigenvalues $1 - \zeta_{11}^j$, whose quotients by π reduce to j. The formal inverse function theorem therefore gives

$$\lambda : H_m \xrightarrow{\sim} H_{m+1} \qquad (m \ge 1). \tag{4}$$

On leading terms, dividing by λ divides the j-th cotangent-dual coordinate by j, after the common factor π has been removed.

To compute $\mu_v(H_1)$, choose smooth sections with distinct residual X-coordinates $0, 2, 3, 4, 8$; at $X = 0$ choose $Y = -b$. The sum of the five Abel maps is a system of formal coordinates, since its differential is a Vandermonde matrix with unit determinant. For a displacement $X \mapsto X + \pi u$, direct expansion gives

$$\frac{Y_{\text{new}} - b}{Y_{\text{old}} - b} \equiv \begin{cases} 1 + \pi^{11}\bar{u}/(4\bar{b}^2) & \bar{X} = 0, \\ 1 & \bar{X} \in \{2, 3, 4, 8\} \end{cases} \pmod{\pi^{12}}.$$

Indeed, at a nonzero residue the two possible terms of valuation 11 in $(X + \pi u)^{11} - X^{11}$ cancel: $(11/\pi^{10})\bar{X}^{10}\bar{u} + \bar{u}^{11} = 0$. All elements of $1 + \pi^{12}\mathcal{O}_k$ are eleventh powers, by logarithm and exponential on $1 + \pi^2\mathcal{O}_k$. The first line generates the nonzero unramified Kummer line. Its nonvanishing is also verified by the local Kummer maps in the computation. Thus $\dim \mu_v(H_1) = 1$.

Let

$$\ell_0(X) = \frac{(X-2)(X-3)(X-4)(X-8)}{(-2)(-3)(-4)(-8)} = 1 + 2X + 2X^2 + X^3 + 9X^4 \in \mathbb{F}_{11}[X].$$

This is the linear functional extracting the displacement at the section with residual coordinate zero, expressed in the differential basis. For a point P near P_0, suppose $v_k(X(P) - X(P_0)) = d \ge 2$. Then $i(P) \in H_d \setminus H_{d+1}$. Put $Q = \lambda^{-(d-1)}i(P) \in H_1$, using the unique formal divisions in (4). Up to a nonzero scalar, its Kummer image is

$$h_{d-1}(8), \qquad h_r(x) = \sum_{j=1}^{5} \ell_{0,j-1}\, x^{j-1} j^{-r}, \tag{5}$$

where $\ell_{0,j-1}$ is the coefficient of X^{j-1} in ℓ_0. This follows by applying the diagonal leading action of successive formal divisions to the Abel tangent vector $(1, 8, 8^2, 8^3, 8^4)/(2\overline{Y(P_0)})$. All nonlinear terms disappear on the associated graded groups. In particular,

$$r \equiv 1 \pmod{10} \quad \Longrightarrow \quad h_r(8) = h_1(8) = 5 \neq 0. \tag{6}$$

4 From the rational-image locus to the formal obstruction

Write $u = 2 - z$, a uniformizer of the completion of K at 11, normalized by $v_K(u) = 1$. The substitutions $x' = (x - 4)/u^2$, $y' = (y + 1/2)/u^3$ give the integral minimal model

$$(y')^2 = (x')^3 + \frac{11}{u^2}(x')^2 + \frac{33}{u^4}x' + \frac{121}{4u^6}.$$

Its discriminant has valuation three. The point $2G$ is formal, with parameter $-x'/y' = 4u$. If $n \neq n_0$, let $k_0 = v_{11}(n - n_0) \ge 2$. Since $n - n_0$ is even, the elliptic formal logarithm, which is an isometry here, gives

$$v_K(t((n - n_0)G)) = 1 + 5k_0.$$

For $q_0 = n_0G - 3s$, the exact computations give

$$v_K(x(q_0) - 4) = 2, \quad v_K(y(q_0) + 1/2) = 3, \quad v_K(\delta_{n_0}) = 2.$$

Thus q_0 is a smooth point on the special fibre of the minimal model, with unit y', and translation by the formal point gives

$$v_K(x(nG - 3s) - x(q_0)) = 3 + 5k_0.$$

Equation (1) now yields

$$v_K(w^{11} - 1) = 1 + 5k_0, \qquad v_K(w - 1) = 5k_0 - 4.$$

For the second equality, first reduce modulo u to get $\bar{w} = 1$. In the binomial expansion the term $11(w-1)$ then has strictly smaller valuation than all other nonzero terms, since $v_K(11) = 5$. At every $v \mid 11$ of F we obtain

$$d = v_k(X(P) - X(P_0)) = 10k_0 - 8 \geq 12, \qquad d - 1 \equiv 1 \pmod{10}. \tag{7}$$

One must also check that the same global choice of square root puts P near P_0 at all three places. This is not automatic from $w \equiv 1$. There is an explicit choice. Put

$$R = (n/2)G + 4s, \qquad R_0 = (n_0/2)G + 4s,$$

so $2R = nG - 3s$. Then

$$v(R) = \frac{(x(R) - \theta)^2 - (3\theta^2 - 2\theta - 7)}{2y(R) + 1} \tag{8}$$

satisfies $v(R)^2 = x(2R) - \theta$. At R_0 this equals the chosen known square root exactly. Since $4 \mid M$, the difference $R - R_0 = ((n - n_0)/2)G$ is formal; its parameter has valuation at least 22 over each F_v. All R_0 have integral minimal coordinates. The computed valuations are

$$4 \leq v_k(x(R_0) - \theta) \leq 6, \quad v_k(2y(R_0) + 1) \leq 8, \quad v_k(v(R_0)) = 2.$$

Translation in the minimal coordinates changes the original x and y coordinates by quantities of valuations at least 26 and 28. Substituting in (8) shows $v(R)/v(R_0) \equiv 1 \pmod{\pi^{20}}$. This fixes the required signs simultaneously. Since $2Y(P_0)$ is a unit, P is in its smooth residue disc, and (7) applies.

Lemma 2. *Under the Selmer injectivity and intersection conditions in (3), no global divisor $D \in J(F)$ can satisfy $D \in \prod_{v|11} H_{d,v}$, with $d \geq 2$, and have a nonzero Kummer image after its $d - 1$ successive formal λ-divisions.*

Proof. Write Q_j for the formal quotient $\lambda^{-j}D$, for $0 \leq j \leq d-1$. Initially $Q_0 = D$ is global, and its local Kummer image is zero because $H_d \subset \lambda H_1$. Injectivity of Selmer localization implies that its global Kummer class is zero. Hence a global R with $\lambda R = Q_0$ exists. Locally, $R = Q_1 + t_v T$, for scalars $t_v \in \mathbb{F}_{11}$. Its local Kummer image lies both in $\mathcal{S}$ and in $U + T_{\mathrm{loc}}$. By (3), the formal contribution is zero and all t_v coincide. Subtracting this global multiple of T shows that Q_1 is global and has zero global Kummer class, the latter again by localization injectivity. Repeating the argument shows that the local Kummer image of Q_{d-1} is zero, as asserted. □

Proof of Proposition 1. If $n \neq n_0$, the construction above gives $D = i(P) \in J(F)$ with depth $d = 10k_0 - 8$ at every place above 11. Equations (5)–(6) show that its final formal quotient has nonzero Kummer image at every such place. This contradicts Lemma 2. □

The global argument is a direct isogeny version of the division-image method in Stoll's Selmer-group Chabauty framework [1, Section 2]. The local formal calculation and the numerical certificates used here are given explicitly above, rather than inferred from a rank inequality.

5 The remaining odd cases and reproducibility

For an odd state the same genus-five descent has $(4-\theta)v^2 = \alpha - \theta + \delta_{n_0} w^{11}$. Its branch field is $B = F(\sqrt{4-\theta})$, of degree 60. The extension B/L is biquadratic, with degree-30 subfields

$$F = L(\sqrt{-11}), \quad L(\sqrt{4-\theta}), \quad L(\sqrt{-11(4-\theta)}).$$

Since 11 does not divide four, the Kummer space of B, unramified outside 2 and 11, is generated by the images of the corresponding spaces of these three subfields. This follows either from the four characters of the Klein group or from $2 = \sum_{|H|=2} N_H - N_{\mathrm{Gal}(B/L)}$ in its group ring. The two new fields have computed class number one under GRH and signatures $(10, 10)$, $(0, 15)$; their absolute discriminants are $2^{30}11^{28}$, $2^{30}11^{27}$. Their unit computations have completed. This supplies a way around the stalled direct degree-60 calculation. The corresponding subfield Kummer spaces have dimensions 19, 22 and 17. Their images should span a space of dimension 34 over B; the attempted combined localization has not yet certified that span, so no odd Selmer bound is reported.

There is a concrete local difference in the odd cases. After adjoining $\sqrt{4-\theta}$, the unit model is again $Y^2 = X^{11} + a^{10}$, but its known point reduces to $(X, Y) = (-1, 0)$, the singular point, at all three places above 11. The extension B/F is ramified quadratic at two places and unramified quadratic at the third; the normalized B-valuation of $Y(P_0)$ is one at all three. These facts are checked in `verify_odd_geometry.m`. Thus the smooth-disc calculation used above cannot simply be reused. A next local calculation should treat the tangent direction at these singular reductions and compare its division images at the two ramified places with the odd Selmer image. No odd-class exclusion is claimed here.

The main scripts, run from the workspace root, are:

```
python3 outputs/continuation/create_descent.py
magma -b NN:=0 TAG:=0 outputs/continuation/compute_selmer.m \
  outputs/continuation/local_formal_test.m
magma -b outputs/continuation/formal_0_data.m \
  outputs/continuation/audit_matrices.m
magma -b outputs/continuation/verify_even_geometry.m
```

Repeat the descent for $n_0 = -2, 2, 4$, using tags `minus2`, `2`, `4`. The four completed logs are `selmer_formal_*.log`. The exact matrix audit is fast and does not rerun number-field arithmetic. The generator script uses the user's licensed Magma package to make a local copy with additional sampling and certificate hooks; the installed package is untouched, and its source is not redistributed. Magma V2.30-1 was used.

The finite sieve input retains the earlier audit scope: its linear algebra and congruences were independently reproduced, and all 21 local tables at eligible primes at most 200 were regenerated. The higher-prime tables remain inherited data. The proof of Proposition 1 itself uses only the indicated congruence, descent equation, and the new Selmer and local computations.

References

[1] M. Stoll, *Chabauty without the Mordell–Weil group*, preprint, December 2017. https://www.mathe2.uni-bayreuth.de/stoll/papers/ratpts-selmer-2016-12-07.pdf.